WCS Upper/Lower A-Arms Simulation Results

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Forces estimation for WCS:

Total Car weight: 293 kg = 2875 N.

Weight distribution: 40/60.

Bump + Turn (Only 2.5g lateral for lower arms WCS) = 4g vertical ; 1.5g lateral.

Rear Tires Forces : Fx = 1293 N ; Fy = 3448 N

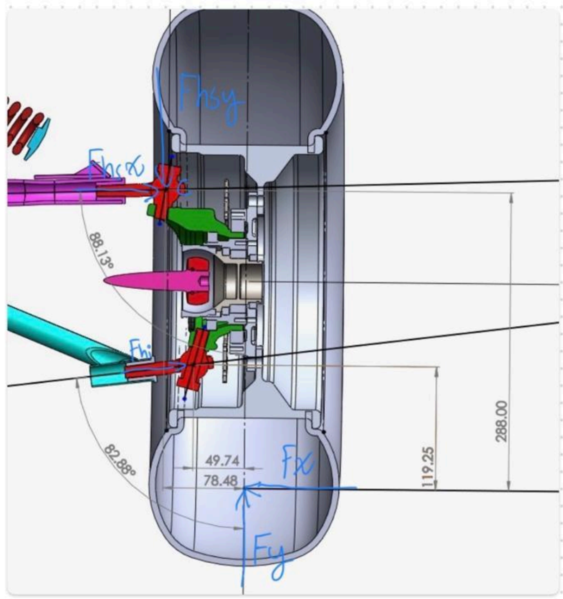
Front Tires Forces (Upper WCS) : Fx = 862.5 N ; Fy = 2300 N

Front Tires Forces (Lower WCS) : Fx = 1437.5 N ; Fy = 575

Forces Distribution:

For the distribution of the forces in each A-Arm the following simplified FBDs were made, without considering intervention of any other component and taking as a rigid body the rim, steering knuckle and wheelhub.

FBD Front Arms:



$F_x = 862.5\,N \quad F_y = 2300\,N$

$$\Sigma M_c = -F_x(288) + F_y(78.48) + F_{hi}(168.75)\sin(82.88) + F_{hi}(28.74)\cos(82.88) = 0$$

$$F_{hi} = \frac{F_x(288) - F_y(78.48)}{167.647} = 404.99\,N$$

$$\Sigma F_x = -F_x + \sin(82.88)F_{hi} + F_{hsx} = 0$$

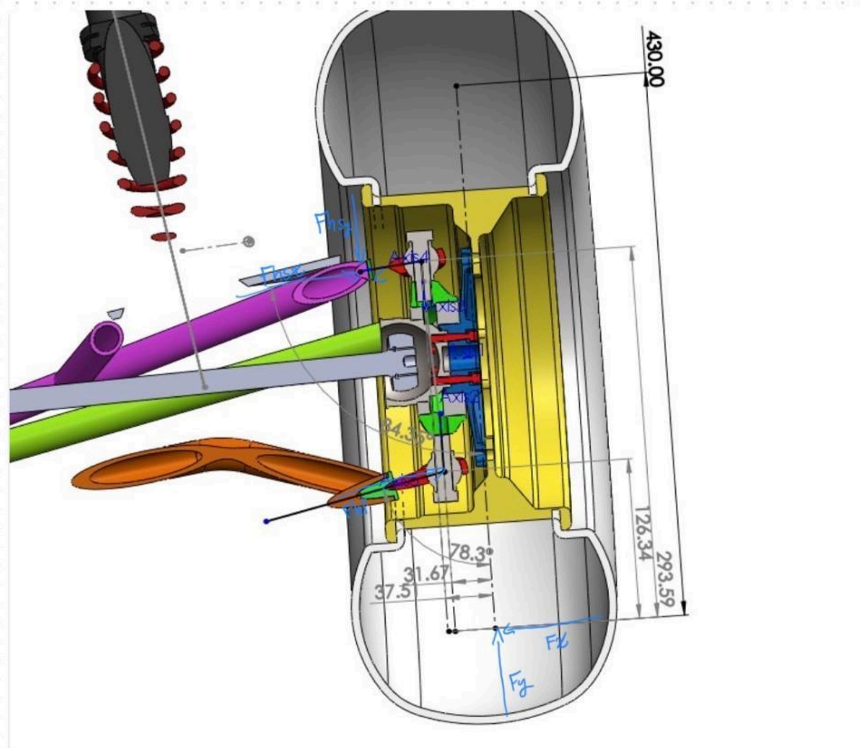
$$F_{hsx} = 485.24$$

$$\Sigma F_y = 2300 - F_{hsy} + F_{hi}\cos(82.88) = 0$$

$$F_{hsy} = 2{,}447.28$$

Force Distribution	Magnitud
Top A-Arm in the X direction (Compression)	485.24 N
Top A-Arm in the Y direction (Upward)	2447.28 N
Lower A-Arm in the X direction (Compression)	4400.6 N

FBD Rear Arms:



$$F_x = 1293 \qquad F_y = 3448$$

$$\sum M_C = F_y(37.51) - F_x(293.59) + F_{hi}\sin(78.3)\,169.25 + F_{hi}\cos(78.3)(5.84) = 0$$

$$F_{hi} = \frac{-F_y(37.51) + F_x(293.59)}{164.96} = 1517.2$$

$$\sum F_x = -F_x + F_{hsx} + F_{hi}\sin(78.3) = 0$$

$$F_x = 932.63$$

$$\sum F_y = F_y - F_{hsy} + F_{hi}\cos(78.3) = 0$$

$$F_{hsy} = 2{,}017.56$$

Force Distribution	Magnitud
Top A-Arm in the X direction (Compression)	932.63 N
Top A-Arm in the Y direction (Upward)	2017.56 N
Lower A-Arm in the X direction (Compression)	3641.05 N

Front Upper A-Arms

Configuration 1:

This configuration represents the baseline design, utilizing an all-steel construction. The main tubular members are modeled in AISI 4130 Chromoly (OD 1'' , ⅛''), while the mounting brackets (ears) are made of AISI 1020 Steel. This setup provides high global stiffness (E = 200 GPa) and excellent yield strength, ensuring minimal deflection and a high Factor of Safety (FOS) under Worst-Case Scenario (WCS) loads. However, it represents the heaviest iteration of the A-Arm.

Configuration 2:

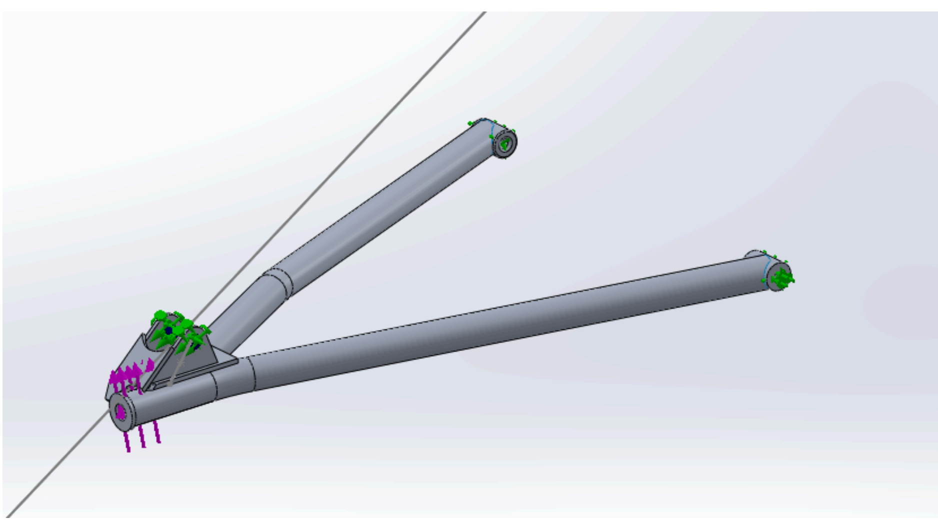
This configuration evaluates a mass-optimization approach by replacing all components (tubes and brackets) with 6061-T6 Aluminum (OD 1'' , ⅛''). While it significantly reduces the unsprung mass of the suspension, the lower elastic modulus (E = 70 GPa) increases global deflection. This higher flexibility significantly lowers the structural Factor of Safety compared to the steel baseline. The shock absorber brackets were thickened to reach the desired FOS in those components, from 3 mm to 6.35 mm.

SetUp:

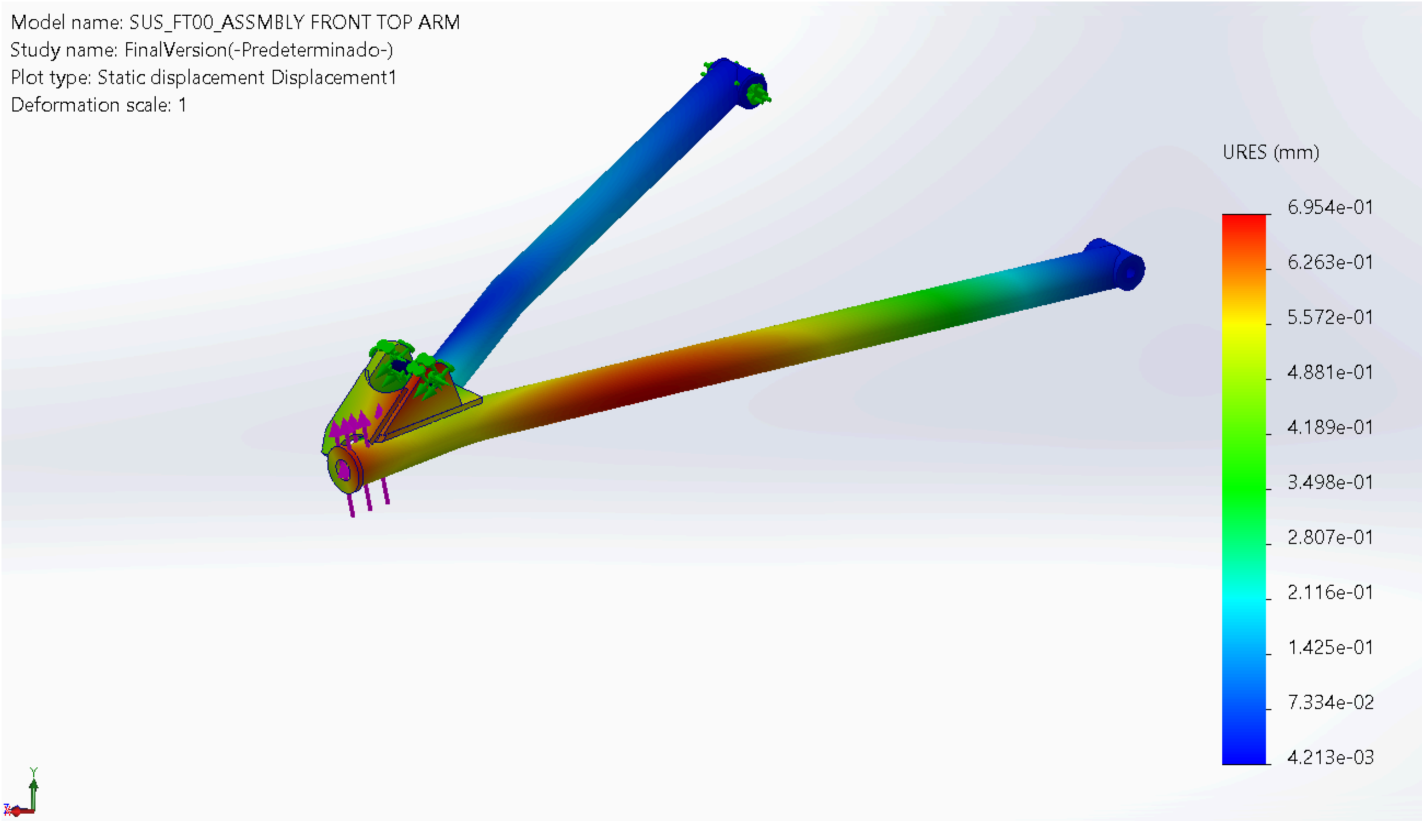
Fixed hinge on bushings rotation axis.

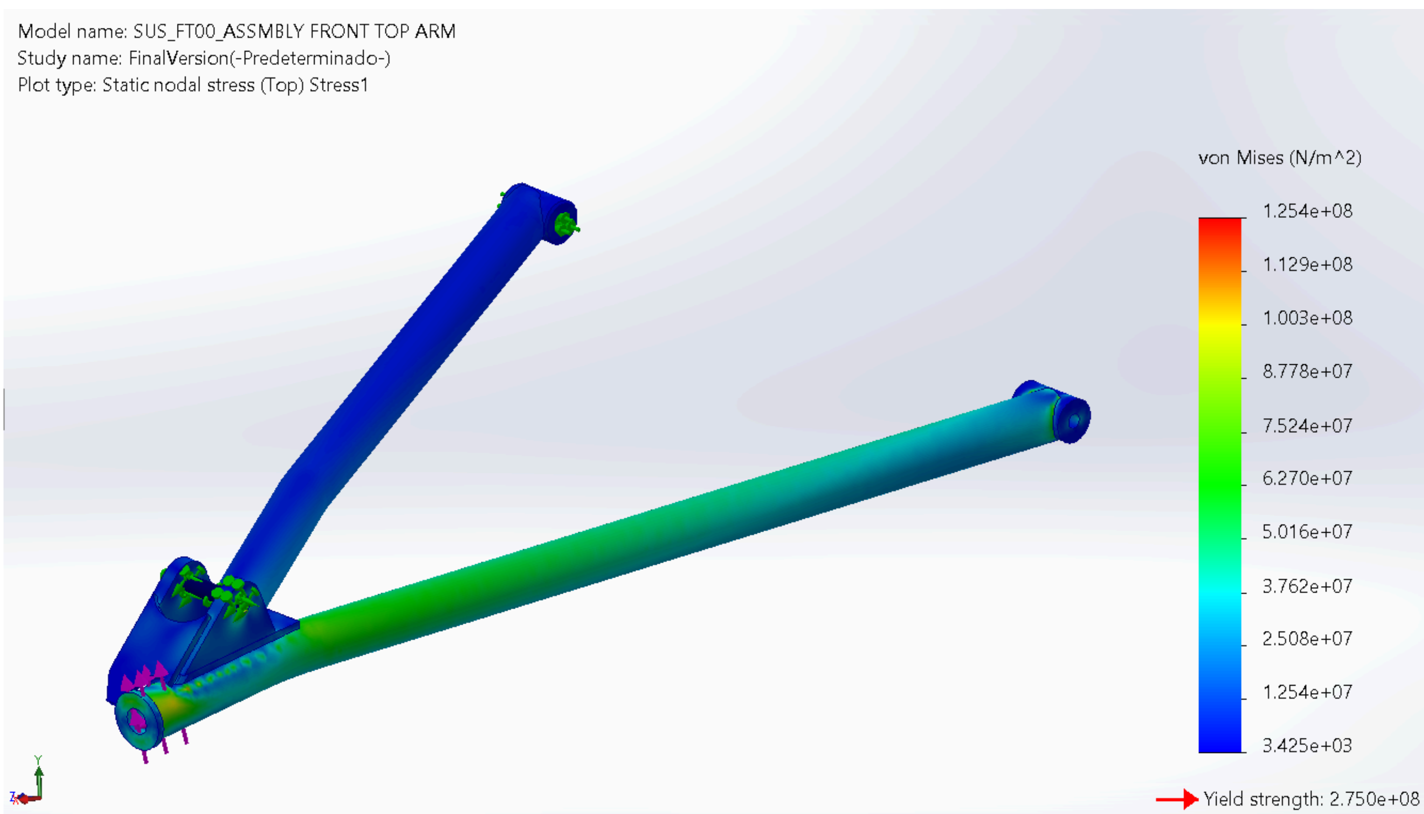
Hinge on shock absorber bearings limited translation through the shock absorber direction (full compression), with bolt connection between the two bearings.

Forces applied in the ball joint bearing connection.



Results:


Model name: SUS_FT00_ASSMBLY FRONT TOP ARM
Study name: FinalVersion(-Predeterminado-)
Plot type: Static displacement Displacement1
Deformation scale: 1
URES (mm)
6.954e-01
6.263e-01
5.572e-01
4.881e-01
4.189e-01
3.498e-01
2.807e-01
2.116e-01
1.425e-01
7.334e-02
4.213e-03


Model name: SUS_FT00_ASSMBLY FRONT TOP ARM
Study name: FinalVersion(-Predeterminado-)
Plot type: Static nodal stress (Top) Stress1
von Mises (N/m^2)
1.254e+08
1.129e+08
1.003e+08
8.778e+07
7.524e+07
6.270e+07
5.016e+07
3.762e+07
2.508e+07
1.254e+07
3.425e+03
Yield strength: 2.750e+08

MATRIZ DE DECISIÓN ESTRUCTURAL

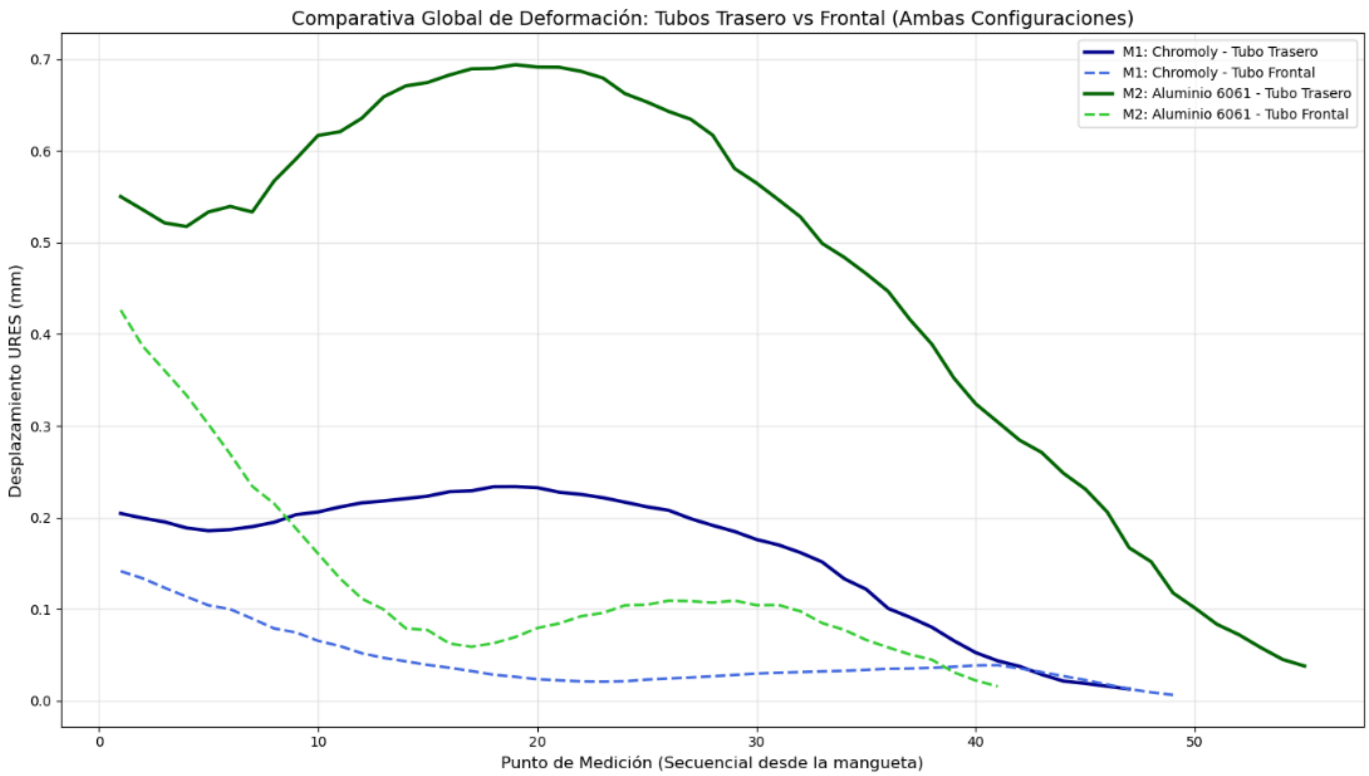
	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Eficiencia_FOS_kg	Status
0	M1_4130_Tubos	110.1	4.18	1.87361	2.230987	Passed
1	M1_AISI1020_Orejas	162.3	2.16	1.87361	1.152855	Passed
2	M2_Aluminio6061_Tubos	106.8	2.57	0.64390	3.991303	Passed
3	M2_Aluminio6061_Orejas	125.4	2.19	0.64390	3.401149	Passed

MATRIZ DE DECISIÓN ESTRUCTURAL AVANZADA

	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Rigidez_K_N_mm	Rigidez_Especifica
0	M1_Chromoly_Tubos	110.1	4.2	1.9	9005.5	4806.5
1	M2_Aluminio_Tubos	106.8	2.6	0.6	3075.7	4776.7

REPORTE FINAL DE OPTIMIZACIÓN Y ELASTOCINEMÁTICA

	Performance Metric	Modelo 1 (Chromoly)	Modelo 2 (Aluminio)	Delta (%)
0	Total Mass (kg)	1.87361	0.6439	-65.6%
1	FOS Min [Tubos]	4.18000	2.5700	-38.5%
2	Global Stiffness K (N/mm)	9005.50000	3075.7000	-65.8%
3	Specific Stiffness ((N/mm)/kg)	4806.50000	4776.7000	-0.6%



Rear Upper A-Arms

Configuration 1:

This configuration represents the baseline design, utilizing an all-steel construction. The main tubular members are modeled in AISI 4130 Chromoly (OD 1'' , ⅛''), as well as the brackets. This setup provides high global stiffness (E = 200 GPa) and excellent yield strength, ensuring minimal deflection and a high Factor of Safety (FOS) under Worst-Case Scenario (WCS) loads. However, it represents the heaviest iteration of the A-Arm.

Configuration 2:

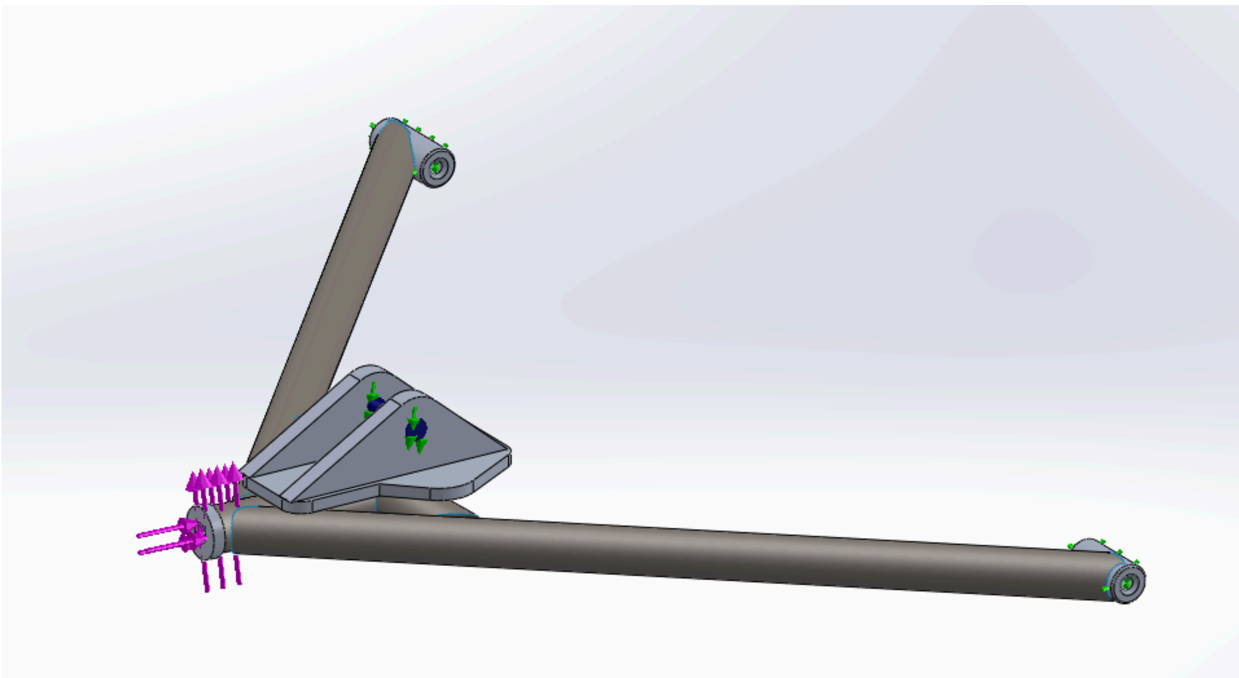
This configuration evaluates a mass-optimization approach by replacing all components (tubes and brackets) with 6061-T6 Aluminum (OD 1'' , ⅛''). While it significantly reduces the unsprung mass of the suspension, the lower elastic modulus (E = 70 GPa) increases global deflection. This higher flexibility significantly lowers the structural Factor of Safety compared to the steel baseline. The shock absorber brackets were thickened to reach the desired FOS in those components, from 3 mm to 9.525 mm.

SetUp:

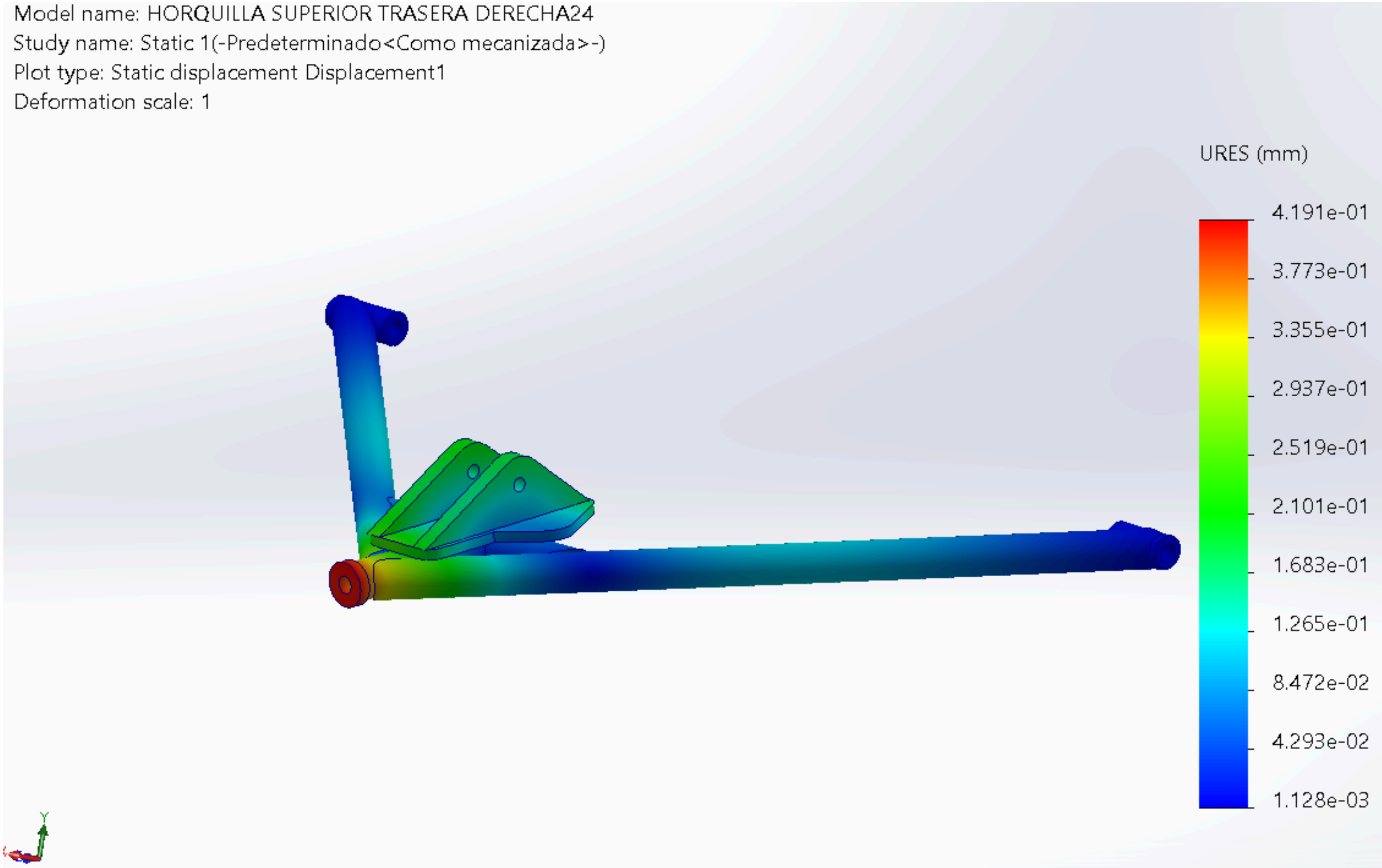
Fixed hinge on bushings rotation axis.

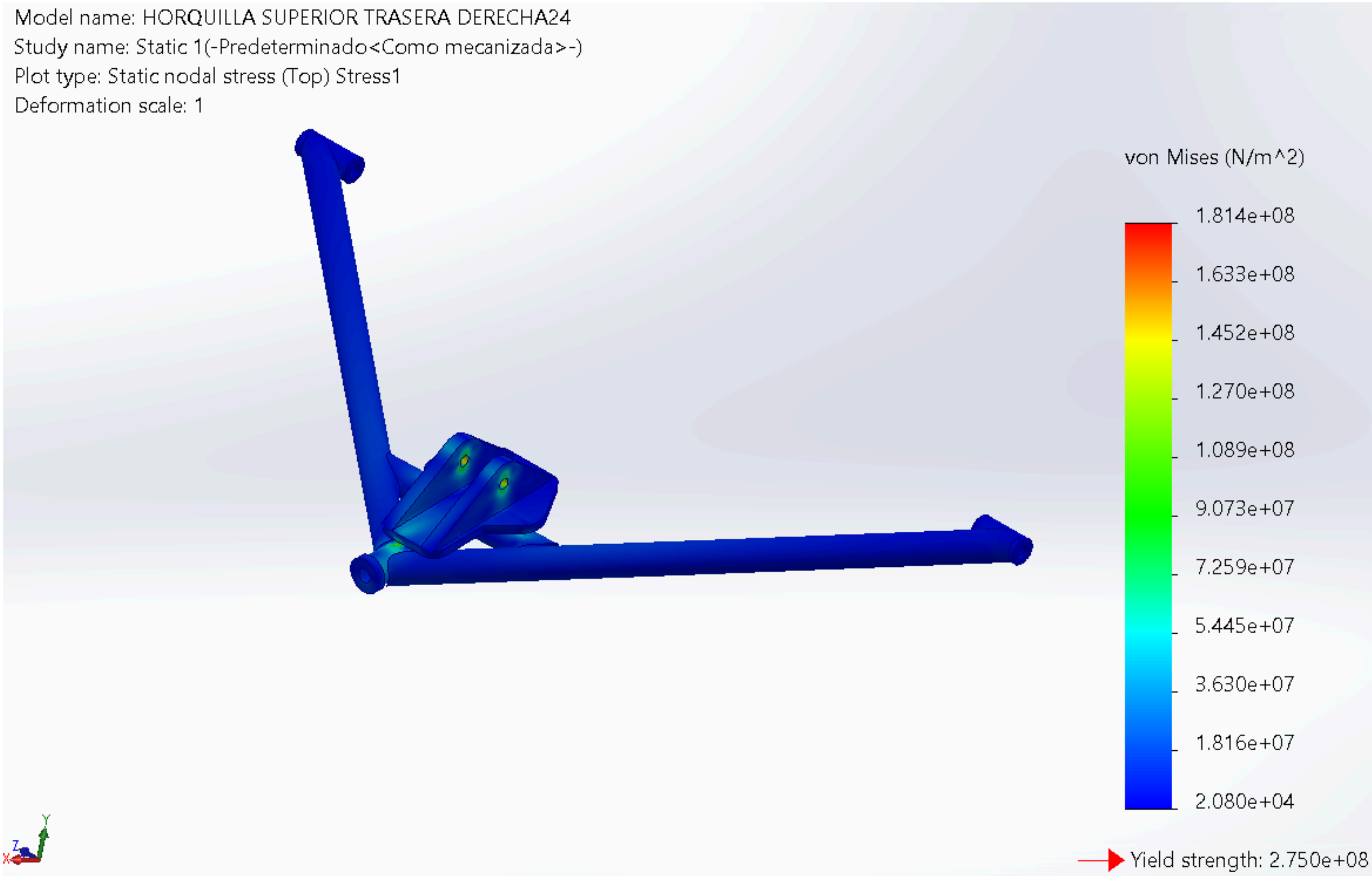
Hinge on shock absorber bearings limited translation through the shock absorber direction (full compression), with bolt connection between the two bearings.

Forces applied in the ball joint bearing connection.



Results:


Model name: HORQUILLA SUPERIOR TRASERA DERECHA24
Study name: Static 1(-Predeterminado<Como mecanizada>-)
Plot type: Static displacement Displacement1
Deformation scale: 1
URES (mm)
4.191e-01
3.773e-01
3.355e-01
2.937e-01
2.519e-01
2.101e-01
1.683e-01
1.265e-01
8.472e-02
4.293e-02
1.128e-03
Y


Model name: HORQUILLA SUPERIOR TRASERA DERECHA24
Study name: Static 1(-Predeterminado<Como mecanizada>-)
Plot type: Static nodal stress (Top) Stress1
Deformation scale: 1
von Mises (N/m^2)
1.814e+08
1.633e+08
1.452e+08
1.270e+08
1.089e+08
9.073e+07
7.259e+07
5.445e+07
3.630e+07
1.816e+07
2.080e+04
Yield strength: 2.750e+08

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MATRIZ DE DECISIÓN ESTRUCTURAL
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	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Eficiencia_FOS_kg	Status
0	M1_4130_Tubos	140.0	3.29	2.713	1.212680	Passed
1	M1_AISI1020_Orejas	109.8	4.19	2.713	1.544416	Passed
2	M2_Aluminio6061_Tubos	148.4	1.85	0.933	1.982851	Failed
3	M2_Aluminio6061_Orejas	181.4	1.52	0.933	1.629153	Failed

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MATRIZ DE DECISIÓN ESTRUCTURAL AVANZADA
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	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Rigidez_K_N_mm	Rigidez_Especifica
0	M1_Chromoly_Tubos	140.0	3.3	2.7	23669.9	8724.6
1	M2_Aluminio_Tubos	148.4	1.8	0.9	5935.5	6361.7

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REPORTE FINAL DE OPTIMIZACIÓN Y ELASTOCINEMÁTICA
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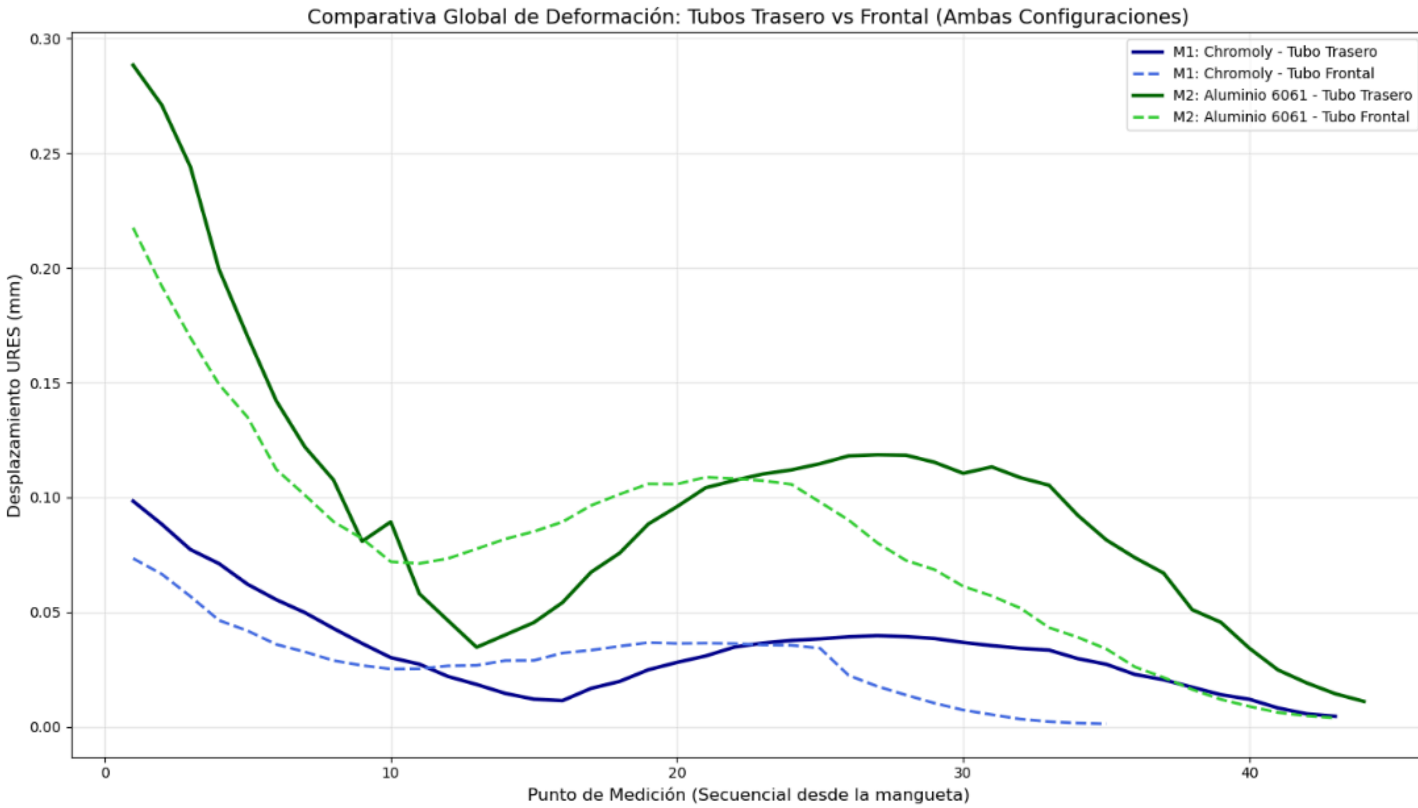
	Performance Metric	Modelo 1 (Chromoly)	Modelo 2 (Aluminio)	Delta (%)
0	Total Mass (kg)	2.713	0.933	-65.6%
1	FOS Min [Tubos]	3.290	1.850	-43.8%
2	Global Stiffness K (N/mm)	23669.900	5935.500	-74.9%
3	Specific Stiffness ((N/mm)/kg)	8724.600	6361.700	-27.1%



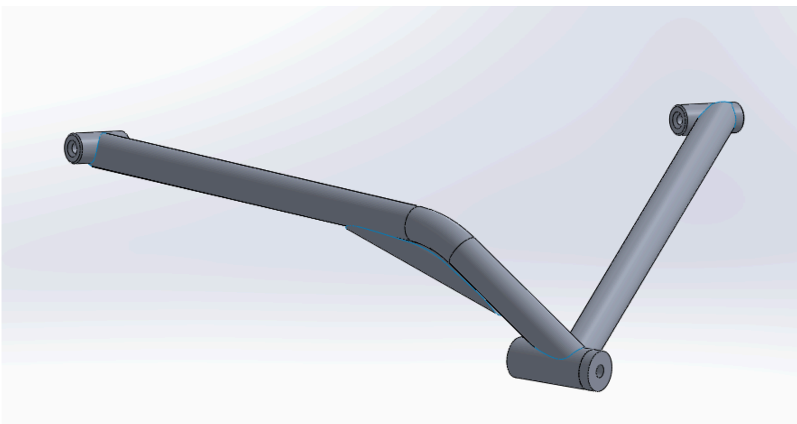
Rear Lower A-Arms

Configuration 1:

This configuration represents the baseline design, utilizing an all-aluminum construction. The main tubular members are modeled in 6061 T6 Aluminum (OD 1'' , ⅛''), as well as the bearings.

Configuration 2:

This configuration maintain the same 6061 T6 Aluminum (OD 1'' , ⅛'') through all the structural members, but with an additional tubular structure in the bend section of the front tube of the A-Arm, as shown below, this adds a a little weight but minimizing the displacement on that tube, as well as helping with the stresses distribution.

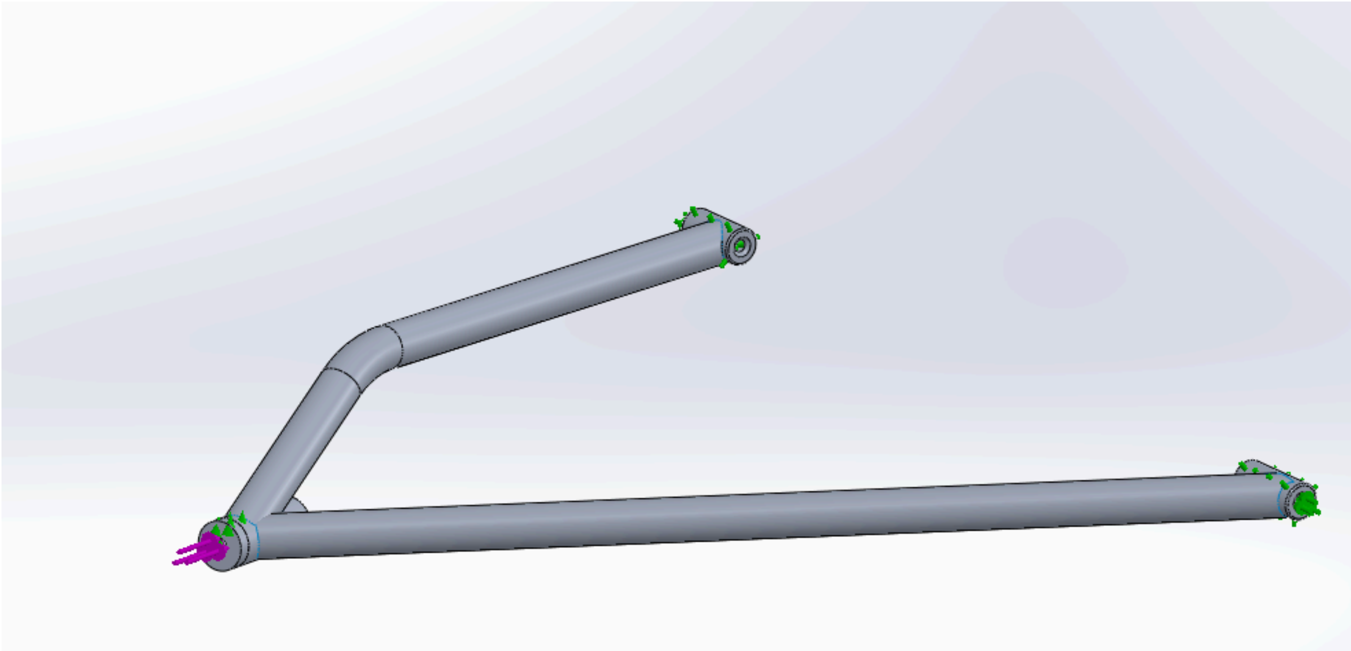


SetUp:

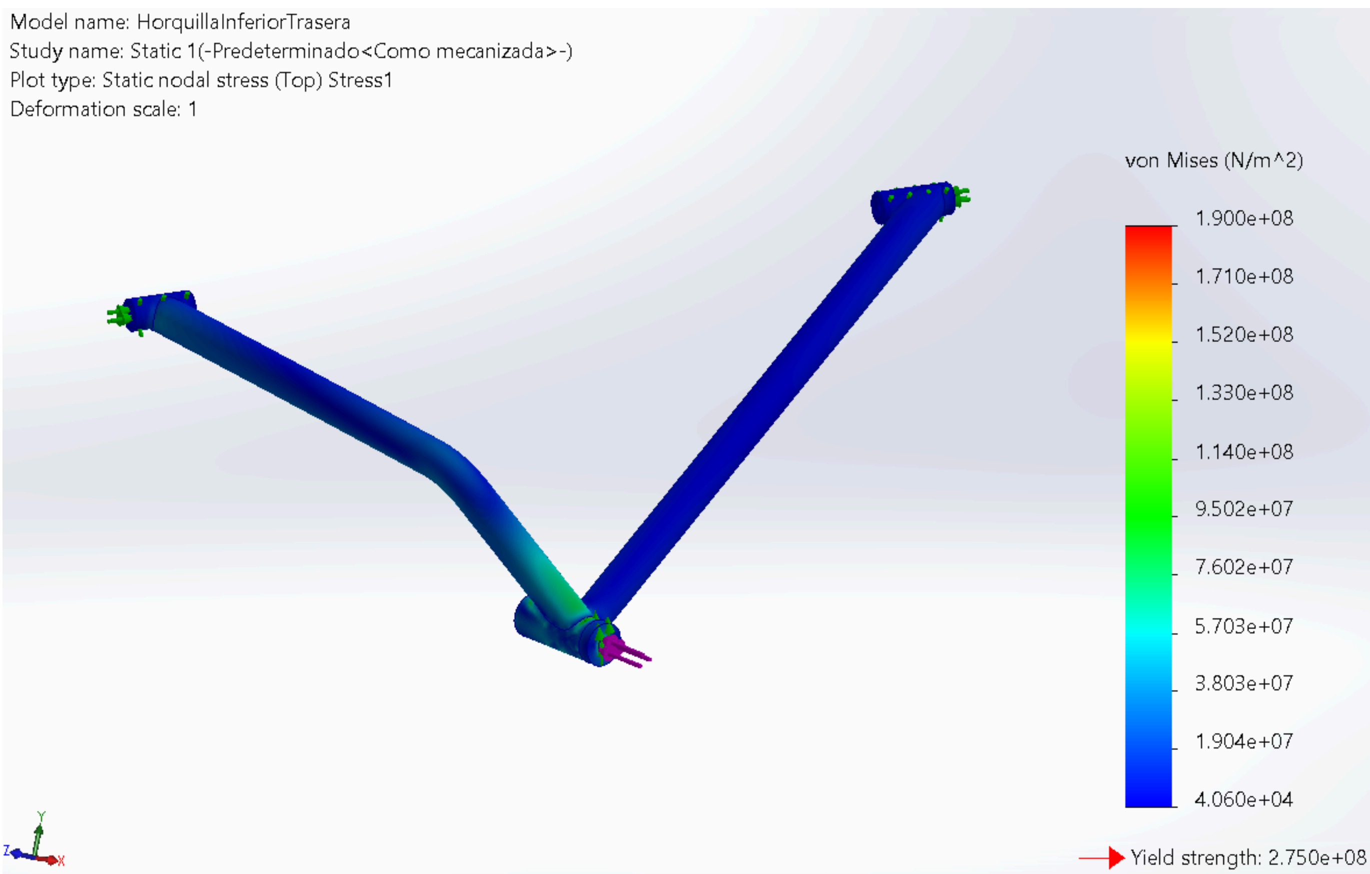
Fixed hinge on bushings rotation axis.

The translation upward (normal to the A-Arms plane) was limited to mimic the steering knuckle movement restriction.

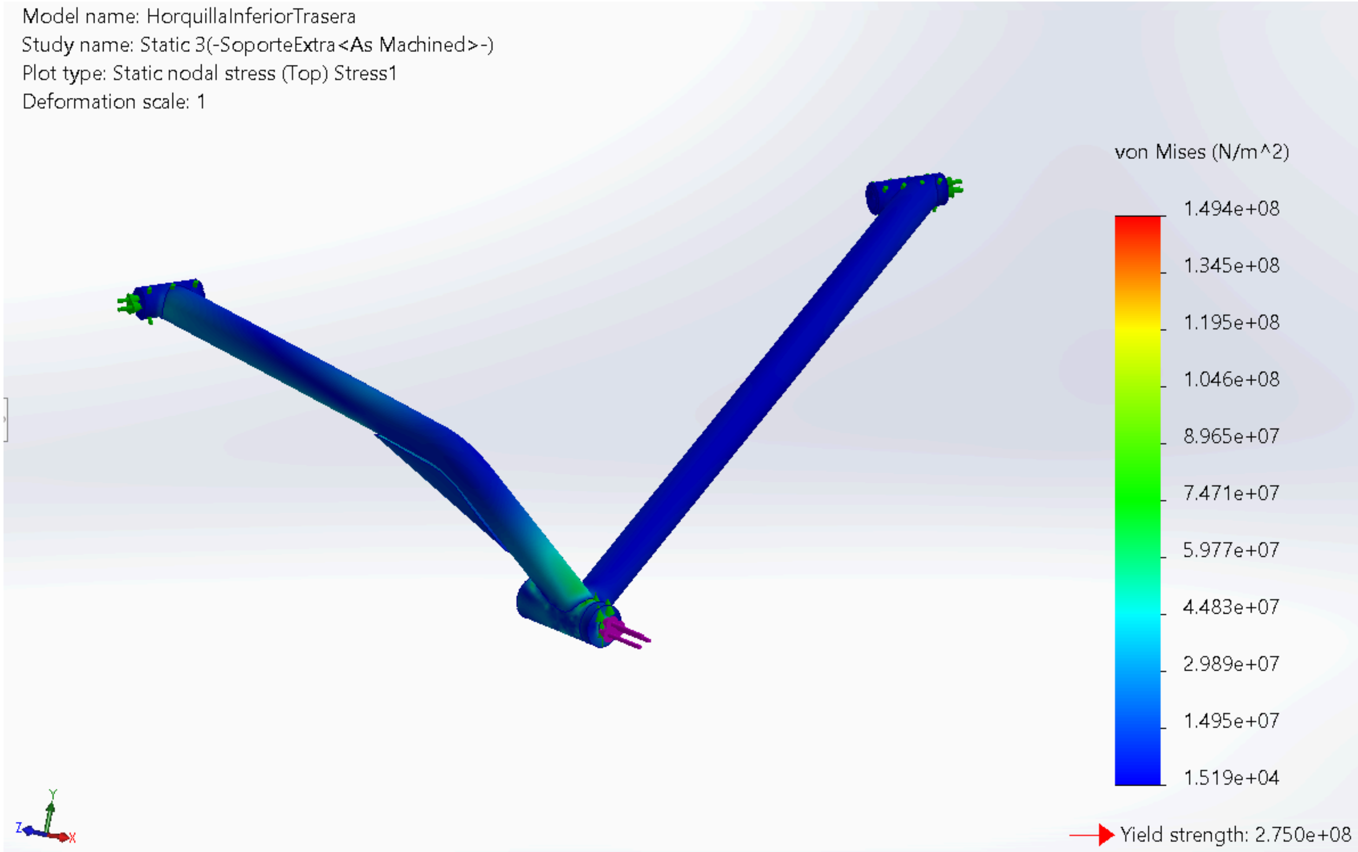
Forces applied in the ball joint bearing connection.

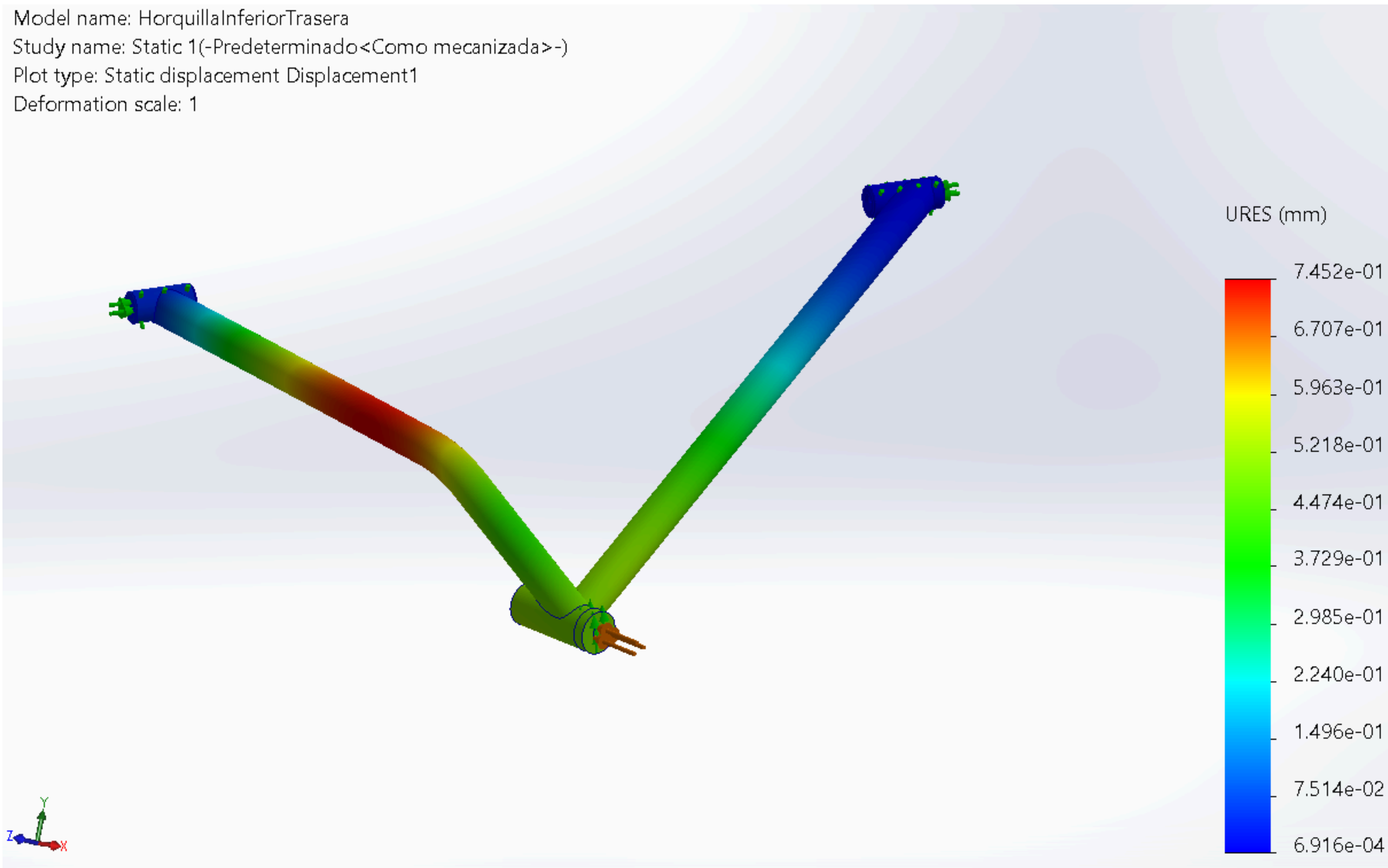


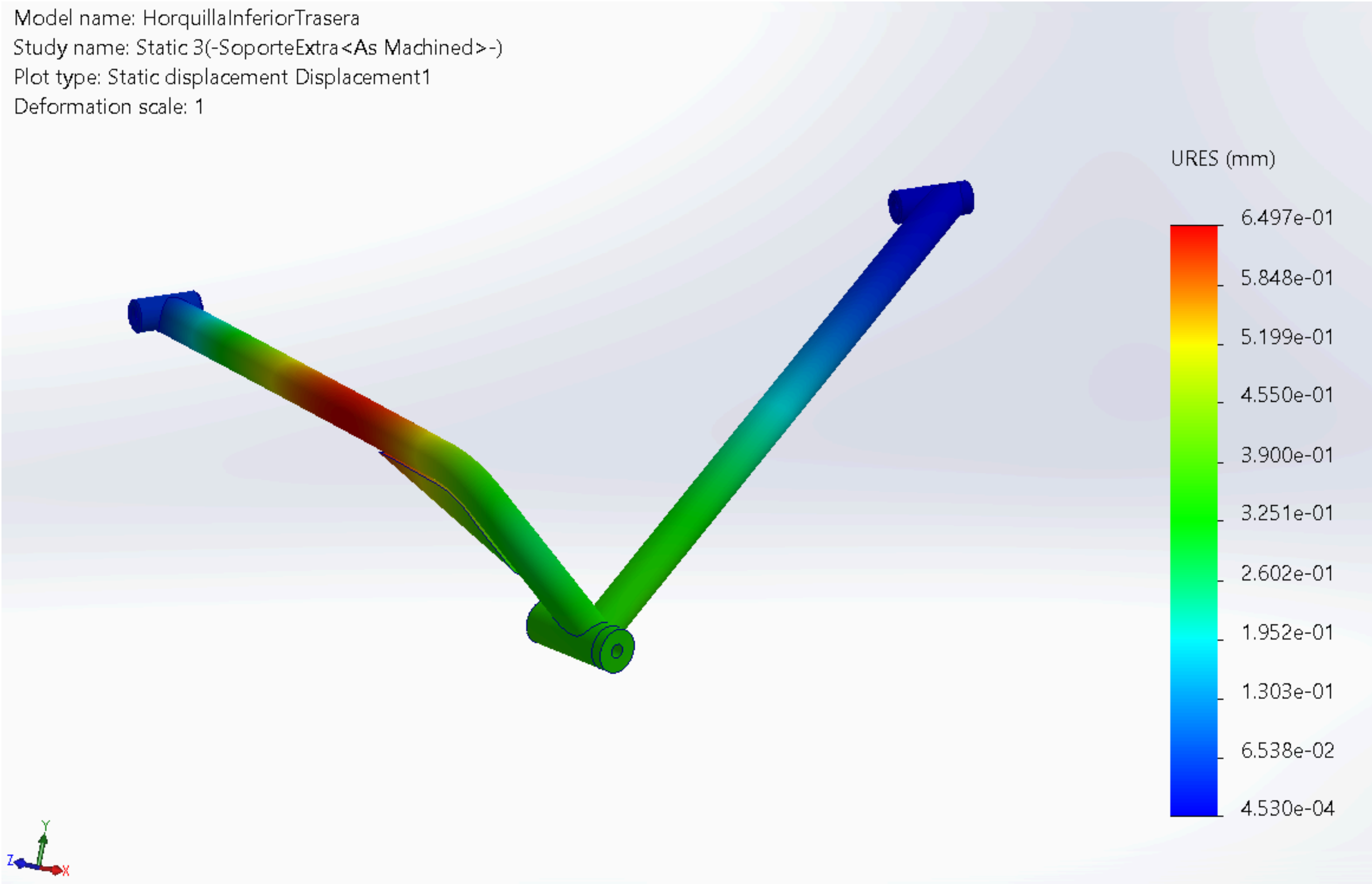
Results:



Model name: HorquillaInferiorTrasera
Study name: Static 1(-Predeterminado<Como mecanizada>-)
Plot type: Static nodal stress (Top) Stress1
Deformation scale: 1
von Mises (N/m^2)
1.900e+08
1.710e+08
1.520e+08
1.330e+08
1.140e+08
9.502e+07
7.602e+07
5.703e+07
3.803e+07
1.904e+07
4.060e+04
Yield strength: 2.750e+08
Y
Z
X


Model name: HorquillaInferiorTrasera
Study name: Static 3(-SoporteExtra<As Machined>-)
Plot type: Static nodal stress (Top) Stress1
Deformation scale: 1
von Mises (N/m^2)
1.494e+08
1.345e+08
1.195e+08
1.046e+08
8.965e+07
7.471e+07
5.977e+07
4.483e+07
2.989e+07
1.495e+07
1.519e+04
Yield strength: 2.750e+08


Model name: HorquillaInferiorTrasera
Study name: Static 1(-Predeterminado<Como mecanizada>-)
Plot type: Static displacement Displacement1
Deformation scale: 1
URES (mm)
7.452e-01
6.707e-01
5.963e-01
5.218e-01
4.474e-01
3.729e-01
2.985e-01
2.240e-01
1.496e-01
7.514e-02
6.916e-04
Y
Z
X


Model name: HorquillaInferiorTrasera
Study name: Static 3(-SoporteExtra<As Machined>-)
Plot type: Static displacement Displacement1
Deformation scale: 1
URES (mm)
6.497e-01
5.848e-01
5.199e-01
4.550e-01
3.900e-01
3.251e-01
2.602e-01
1.952e-01
1.303e-01
6.538e-02
4.530e-04
Y
Z
X

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MATRIZ DE DECISIÓN ESTRUCTURAL
```

	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Eficiencia_FOS_kg	Status
0	M1_Aluminio6061_Tubos	78.76	3.49	0.643720	5.421612	Passed
1	M1_Aluminio6061_Bujes	190.00	1.45	0.643720	2.252532	Failed
2	M2_Aluminio6061_Tubos	73.00	3.77	0.680029	5.543883	Passed
3	M2_Aluminio6061_Bujes	149.40	1.84	0.680029	2.705768	Failed

```
MATRIZ DE DECISIÓN ESTRUCTURAL AVANZADA
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	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Rigidez_K_N_mm	Rigidez_Especifica
0	M1_Aluminio_Tubos	78.8	3.5	0.6	3086.4	4794.7
1	M2_AluminioExtraSupport_Tubos	73.0	3.8	0.7	3540.1	5205.8

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REPORTE FINAL DE OPTIMIZACIÓN Y ELASTOCINEMÁTICA
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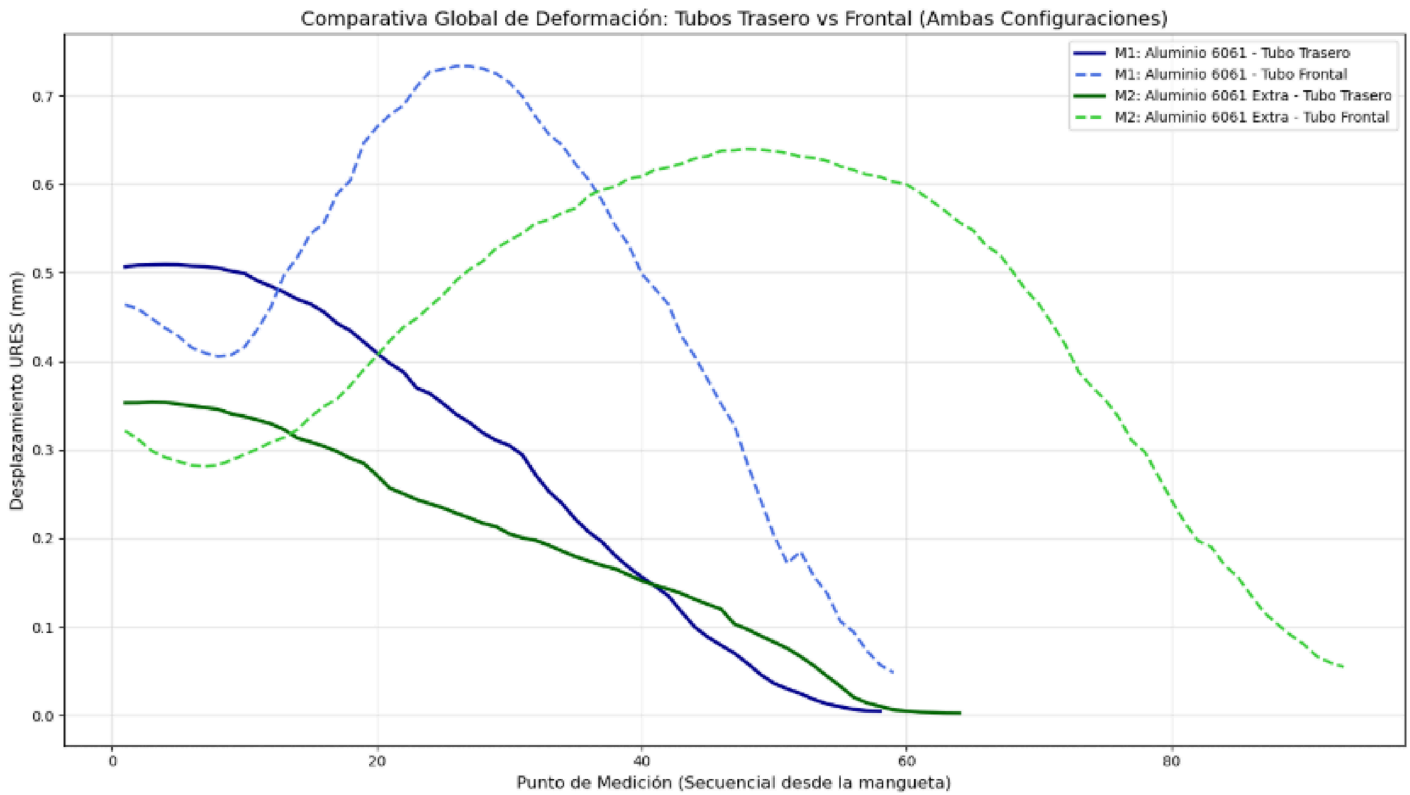
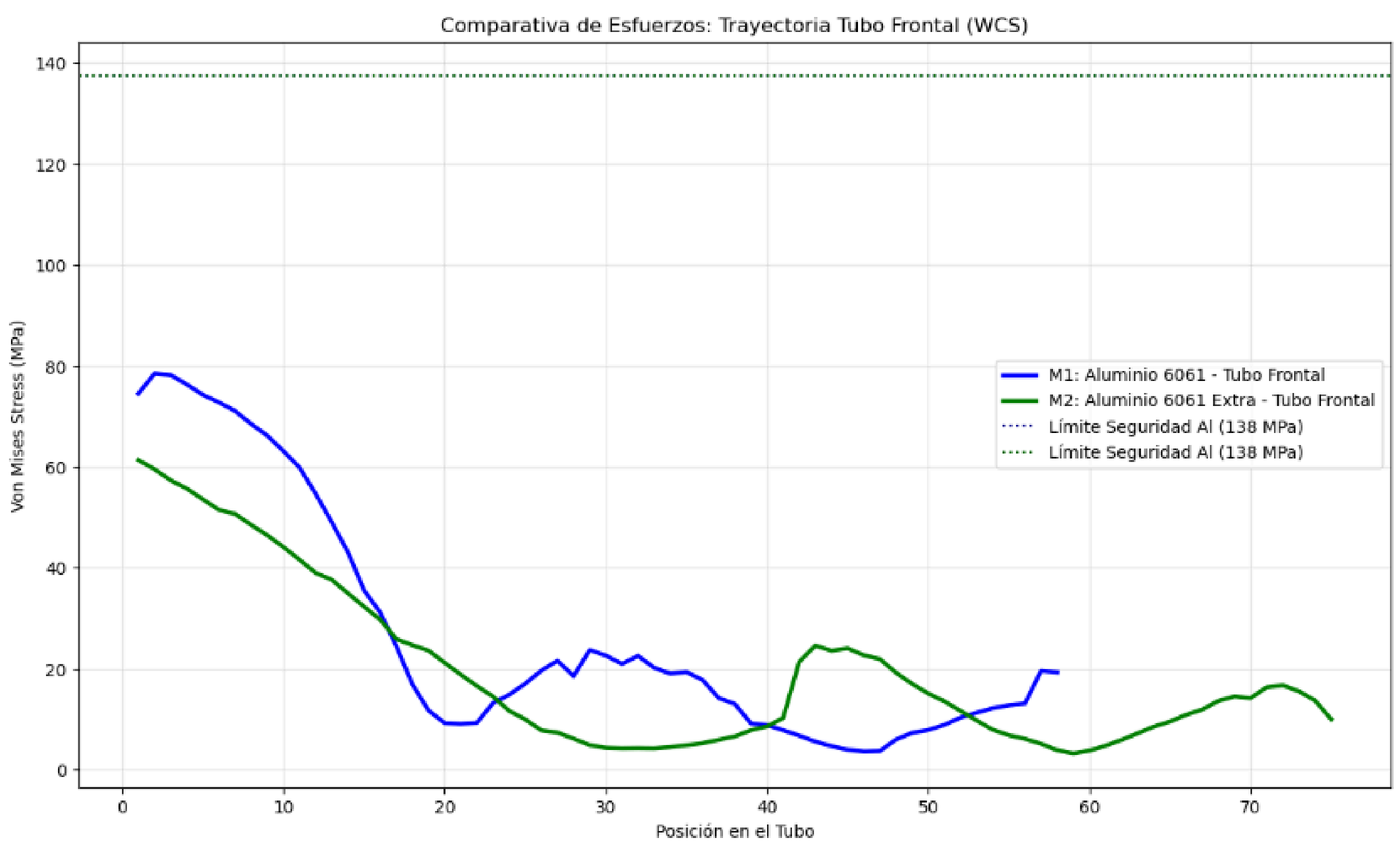
	Performance Metric	Modelo 1 (Aluminio)	Modelo 2 (AluminioExtra)	Delta (%)
0	Total Mass (kg)	0.64372	0.680029	+5.6%
1	FOS Min [Tubos]	3.49000	3.770000	+8.0%
2	Global Stiffness K (N/mm)	3086.40000	3540.100000	+14.7%
3	Specific Stiffness ((N/mm)/kg)	4794.70000	5205.800000	+8.6%


Comparativa Global de Deformación: Tubos Trasero vs Frontal (Ambas Configuraciones)
M1: Aluminio 6061 - Tubo Trasero
M1: Aluminio 6061 - Tubo Frontal
M2: Aluminio 6061 Extra - Tubo Trasero
M2: Aluminio 6061 Extra - Tubo Frontal
Desplazamiento URES (mm)
Punto de Medición (Secuencial desde la mangueta)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0
20
40
60
80


Comparativa de Esfuerzos: Trayectoria Tubo Frontal (WCS)
M1: Aluminio 6061 - Tubo Frontal
M2: Aluminio 6061 Extra - Tubo Frontal
Límite Seguridad Al (138 MPa)
Límite Seguridad Al (138 MPa)
Von Mises Stress (MPa)
Posición en el Tubo
0
20
40
60
80
100
120
140
0
10
20
30
40
50
60
70

Front Lower A-Arm

Configuration 1:

This configuration represents the baseline design, utilizing an all-steel construction. The main tubular members are modeled in AISI 4130 Chromoly (OD 1.25'' , ¼''), while the bearings are made of AISI 1020 Steel. This setup provides high global stiffness (E = 200 GPa) and excellent yield strength, ensuring minimal deflection and a high Factor of Safety (FOS) under Worst-Case Scenario (WCS) loads. However, it represents the heaviest iteration of the A-Arm.

Configuration 2:

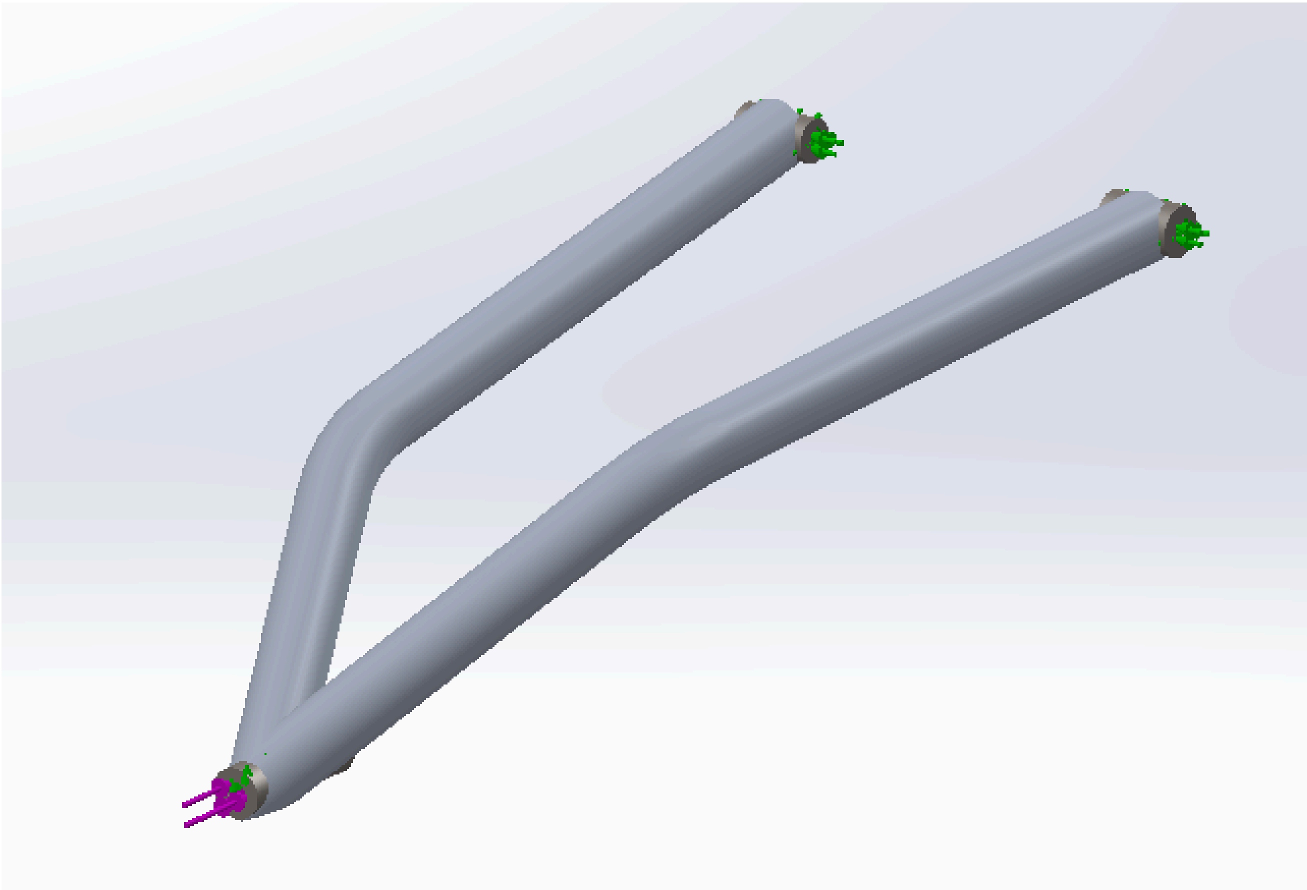
This configuration evaluates a mass-optimization approach by replacing all components (tubes and bearings) with 6061-T6 Aluminum (OD 1.25'' , ¼''). While it significantly reduces the unsprung mass of the suspension, the lower elastic modulus (E = 70 GPa) increases global deflection.

SetUp:

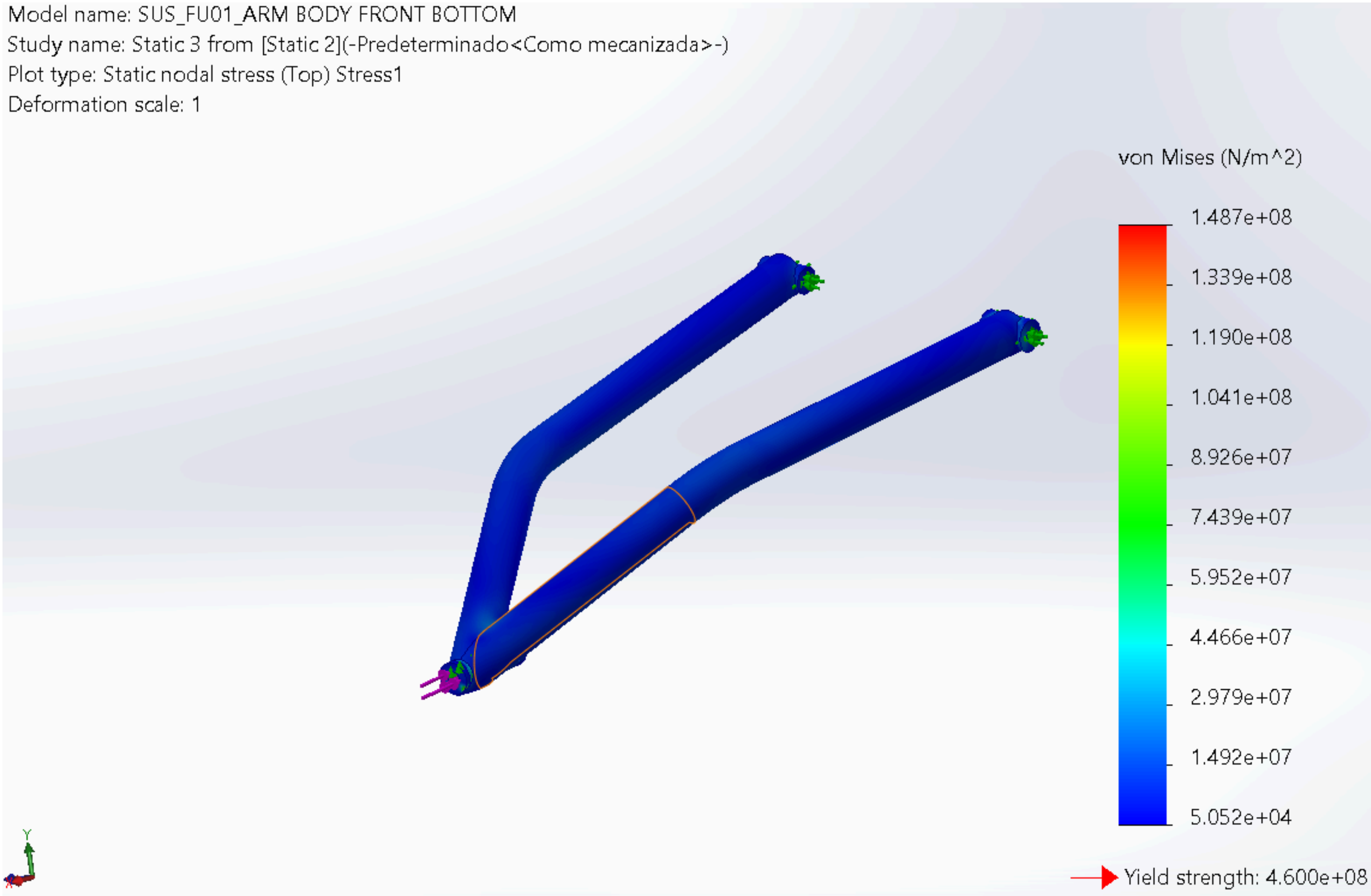
Fixed hinge on bushings rotation axis.

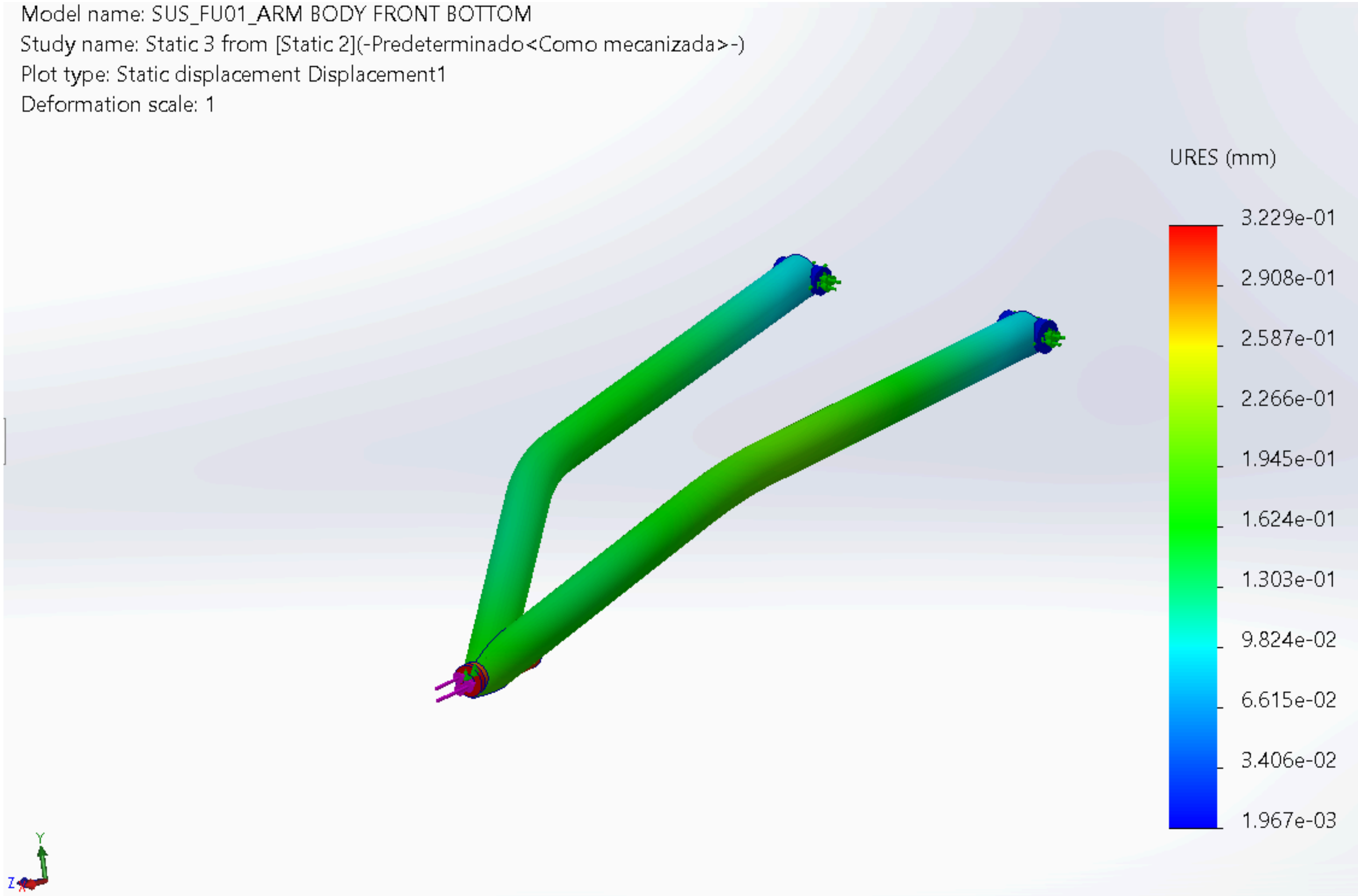
The translation upward (normal to the A-Arms plane) was limited to mimic the steering knuckle movement restriction.

Forces applied in the ball joint bearing connection.



Results:


Model name: SUS_FU01_ARM BODY FRONT BOTTOM
Study name: Static 3 from [Static 2](-Predeterminado<Como mecanizada>-)
Plot type: Static nodal stress (Top) Stress1
Deformation scale: 1
von Mises (N/m^2)
1.487e+08
1.339e+08
1.190e+08
1.041e+08
8.926e+07
7.439e+07
5.952e+07
4.466e+07
2.979e+07
1.492e+07
5.052e+04
Yield strength: 4.600e+08


Model name: SUS_FU01_ARM BODY FRONT BOTTOM
Study name: Static 3 from [Static 2](-Predeterminado<Como mecanizada>-)
Plot type: Static displacement Displacement1
Deformation scale: 1
URES (mm)
3.229e-01
2.908e-01
2.587e-01
2.266e-01
1.945e-01
1.624e-01
1.303e-01
9.824e-02
6.615e-02
3.406e-02
1.967e-03
Y
Z
X

MATRIZ DE DECISIÓN ESTRUCTURAL

	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Eficiencia_FOS_kg	Status
0	M1_4130_Tubos	37.68	12.21	1.602	7.621723	Passed
1	M1_AISI1020_Bujes	148.70	2.36	1.602	1.473159	Passed
2	M2_Aluminio6061_Tubos	37.89	7.26	0.551	13.176044	Passed
3	M2_Aluminio6061_Bujes	147.80	1.86	0.551	3.375681	Failed

MATRIZ DE DECISIÓN ESTRUCTURAL AVANZADA

	ID	Max_Stress_MPa	FOS_Min	Masa_kg	Rigidez_K_N_mm	Rigidez_Especifica
0	M1_Chromoly_Tubos	37.7	12.2	1.6	11408.7	7121.6
1	M2_Aluminio_Tubos	37.9	7.3	0.6	3822.5	6937.4

REPORTE FINAL DE OPTIMIZACIÓN Y ELASTOCINEMÁTICA

	Performance Metric	Modelo 1 (Chromoly)	Modelo 2 (Aluminio)	Delta (%)
0	Total Mass (kg)	1.602	0.551	-65.6%
1	FOS Min [Tubos]	12.210	7.260	-40.5%
2	Global Stiffness K (N/mm)	11408.700	3822.500	-66.5%
3	Specific Stiffness ((N/mm)/kg)	7121.600	6937.400	-2.6%